CSCI 4511W Writing Assignment 2

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1 What problem are the authors trying to solve?

A* is theoretically cost optimal[1], but its cost optimality is based on a single run in a theoretical environment. Real applications of A* are likely to be used in widely varied environments where knowledge retention of previous runs is helpful. This is the case for Adaptive A*. Adaptive A*'s authors intended the algorithm to improve after each subsequent search episode through learning about the nodes it expanded to completion so that heuristics would become more accurate[2].

2 What are the novel contributions stated in the paper by the authors?

As opposed to the original A* algorithm which operates within the limited bounds defined by an initial state, a heuristic, and goal states within a state space[3], the Adaptive A* algorithm persists throughout many search episodes with different initial states and constantly improving heuristics. Each node that was expanded to the goal is used in the heuristic for the next generation of searches that expend the same node. The heuristics improve through being monotonically nondecreasing and becoming more accurate as more nodes are expanded[2].

3 For what type of problems is the proposed algorithm appropriate?

This algorithm is best suited for searches in a state space that doesn't change and goal states don't change. Searches can start from different initial points. The searches and heuristics should improve over time as search episodes are repeated with different initial states and more and more nodes are expanded towards the goal. This algorithm is also useful for state spaces that have constraints that make the heuristic prediction harder. After the first expansion of nodes, subsequent expansions of those nodes will be better informed of constraints that it might run into and have a more accurate heuristic.

4 Do the authors validate their approach experimentally or theoretically? If the validation is done experimentally, are the experimental results sufficient to demonstrate the power of the algorithm?

The authors of the paper validate their approach experimentally by showing how an agent using traditional forward A* would navigate a state space compared to an agent using Adaptive A*. Their experiment shows that while the forward A* searches started each search using the Manhattan distance heuristic, adaptive A* had the advantage of using the updated h-values. Because of this, Adaptive A*'s second exploration (shown on the bottom right of the below graphic) did not have to re-expand the cells that went in the wrong direction (bottom left three cells) whereas forward A* searches still expanded those. Adaptive A* did not have to expand those because their heuristic values came from the previous search episode and took into account the black barrier.

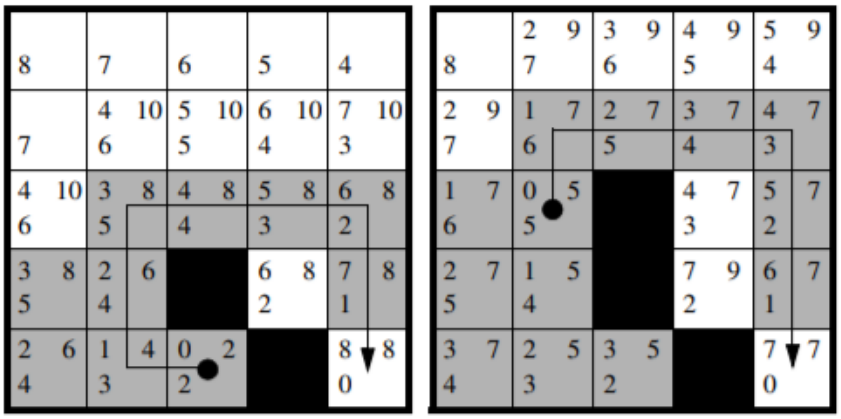


Figure 1: Forward A* Searches

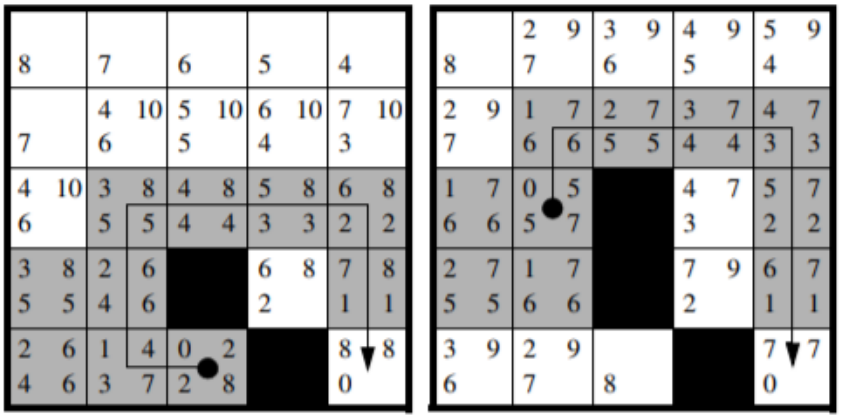


Figure 2: Adaptive A*

[2]

Secondly, results are also demonstrated in the table below where it is shown that Adaptive A* consistently expands less cells than A* (Adaptive A* with binary heap expanded 78.35% of A* and Adaptive A* with buckets expanded 77.36% of A*) and takes less planning time[2].

Expanded Cells		Planning Time (μsec)	
μ	(σ of μ)	μ	(σ of μ)
A* with Binary Heap			
53,084.27	(1,229.42)	16,514.75	(357.13)
Adaptive A* with Binary Heap			
41,593.55	(735.96)	14,355.81	(262.40)
D* Lite with Binary Heap			
11,416.37	(116.94)	8,505.72	(88.01)
Adaptive A* with Buckets			
41,063.69	(541.11)	7,051.12	(96.86)

Table 1: Experiments in Random Mazes

[2]

5 Is the paper clearly written and easy to follow? is the algorithm described clearly so it is possible to implement it and replicate the results?

The paper's intuition was clearly described. While its implementation is not described in depth, it mentions that Adaptive A* is mostly just A* with a special updating mechanism to the heuristic and keeping track of "closed" nodes which are the ones that have been expanded to the goal in previous searches. The results were replicated and it was confirmed that Adaptive A* has better performance than traditional A* in another paper that applied Adaptive A*[4].

6 What did you find most interesting in the paper?

I found the selection process for a updating h(n) to be interesting. I had not really thought about how the expanded node cost would relate to h(n) for a subsequent search before. Other searches that improve over A* include D* Lite which was also conceived by Sven Koenig and Maxim Likhachev in 2002. D* also kept information about paths generated so that paths could simply be updated when an obstacle was found[5].

References

[1] P. E. Hart, N. J. Nilsson, and B. Raphael, "A formal basis for the heuristic determination of minimum cost paths," *IEEE Transactions on Systems Science and Cybernetics SSC4*, 1968.

[2] S. Koenig and M. Likhachev, "Adaptive a*," in *Proceedings of the International Conference on Autonomous Agents and Multi-Agent Systems*, 2005.

[3] S. Russell and P. Norvig, *Artificial Intelligence, A Modern Approach (Fourth Edition)*. New Jersey: Pearson, 2021.

[4] X. Zhao, N. Qin, J. Huang, and Y. Miao, "The application of adaptive a-star algorithm in layout of spatial pipeline," in *2020 IEEE 9th Data Driven Control and Learning Systems Conference (DDCLS)*, IEEE, 2020.

[5] CSMinute, "Intro to path planning: D* lite vs. a*." https://www.youtube.com/watch?v=skK-3UfcXW0, 2015.